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A large, high-resolution image of the Earth as seen from space, showing the Western Hemisphere. The Earth is a vibrant blue and green sphere against a dark, star-filled background. A thin white rectangular border is superimposed over the center of the Earth, framing the main title text.

COMP-2032
**Introduction to
Image Processing**

Lecture 1B
Digital Images and Point Processes



Learning Outcomes

IDENTIFY

1. Digital Images Formation
2. Acquiring & Using Colour Images
3. Colour spaces
4. Intensity Transform





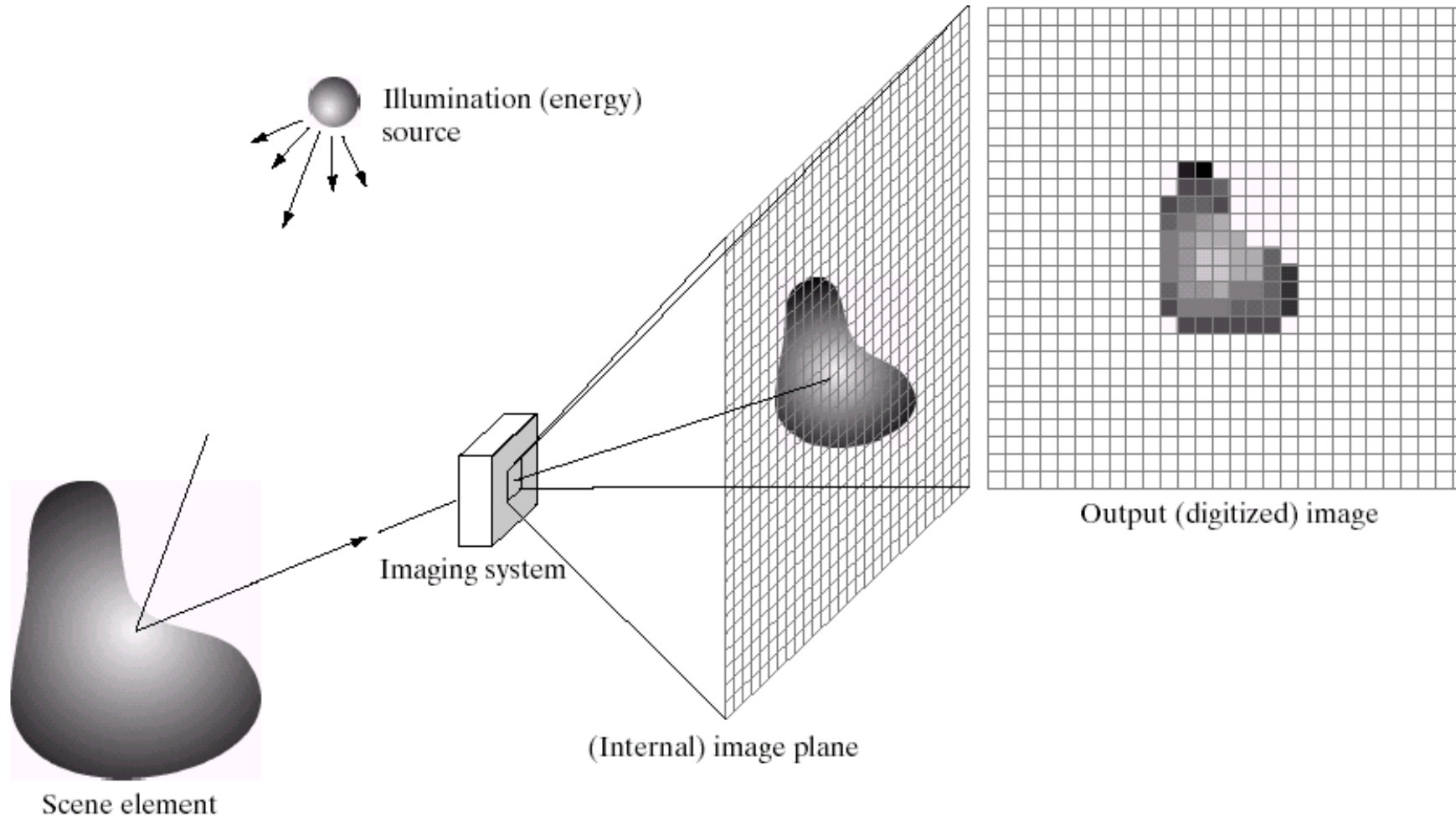
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Digital Image Formation



Image Formation & Acquisition



ACK: Prof. Tony Pridmore, UNUK



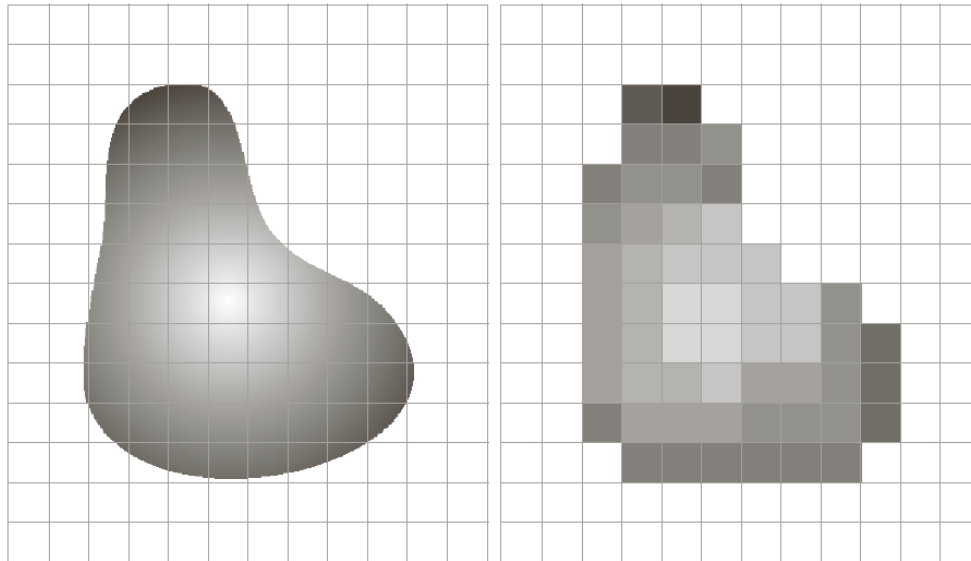
Sampling & Quantisation

Sampling

Digitisation of the **spatial coordinates**

Digitisation of the **light intensity function**

Quantisation



Sampling determines
spatial resolution,
quantisation determines
grey level, colour or
radiometric resolution



Sampling



How many samples to take? i.e., how many pixels in the image?

- Samples must be taken at a rate that is twice the frequency of the highest frequency component to be reconstructed
- Under-sampling: sampling at a rate that is too coarse, i.e., is below the Nyquist rate

Nyquist Rate



Artefacts that result from under-sampling

Aliasing



Aliasing

- Aliasing occurs when two signals (images) become indistinguishable when sampled
- In our case the two signals are the true image (*the image that would be seen if there were no quantisation*) and the one reconstructed by the human vision system from a sampled image



Adequately Sampled Signal



Aliased Signal Due to Undersampling



Quantisation

How many grey levels to store?

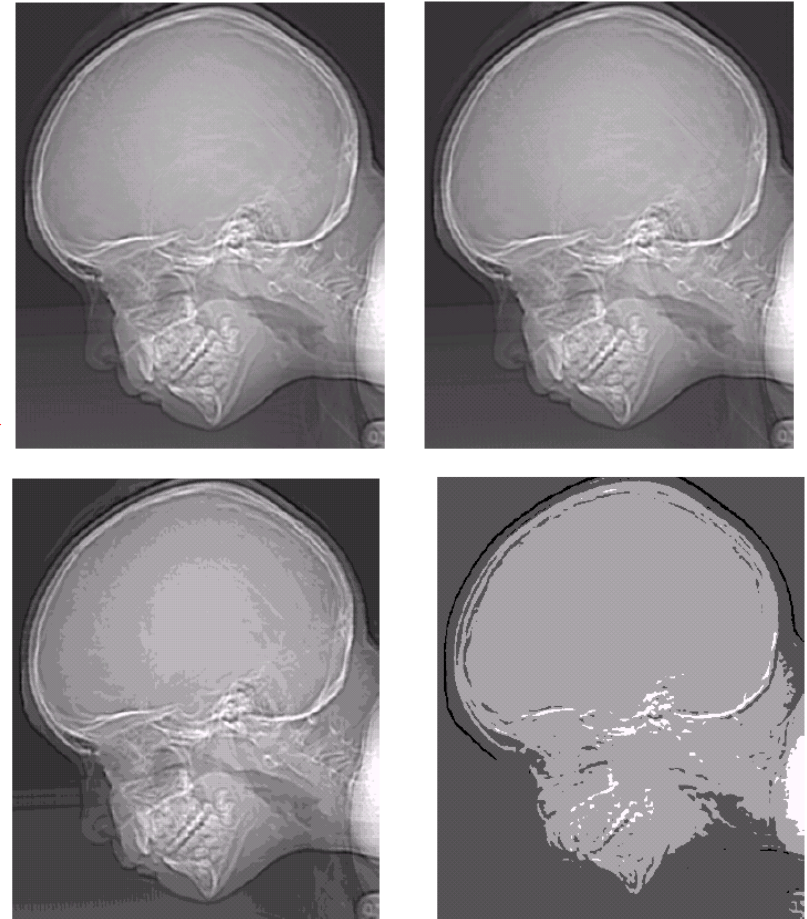
?

- Determines the number of levels of colour/intensity to be represented at each pixel

256, 64, 16, 4

- Sampling and quantisation occur naturally during image acquisition, but can also be applied to existing images

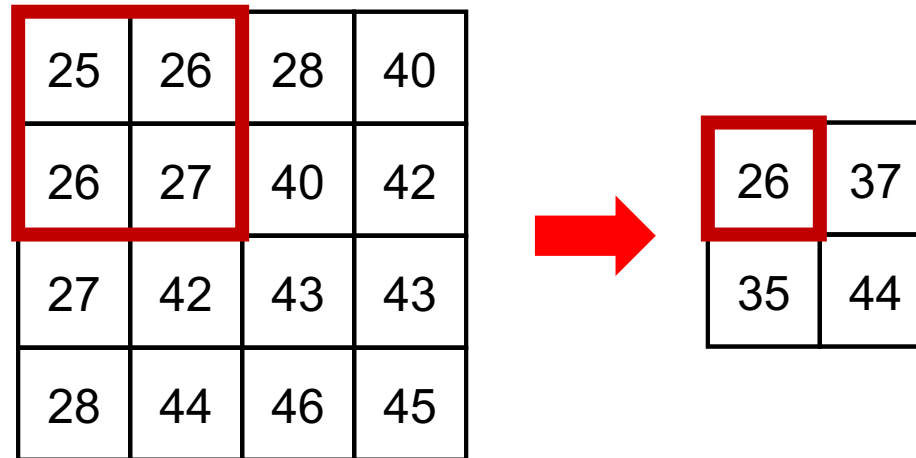
E.g., during resizing & compression





Re-Sampling & Re-Sizing

- Most basic form of image processing

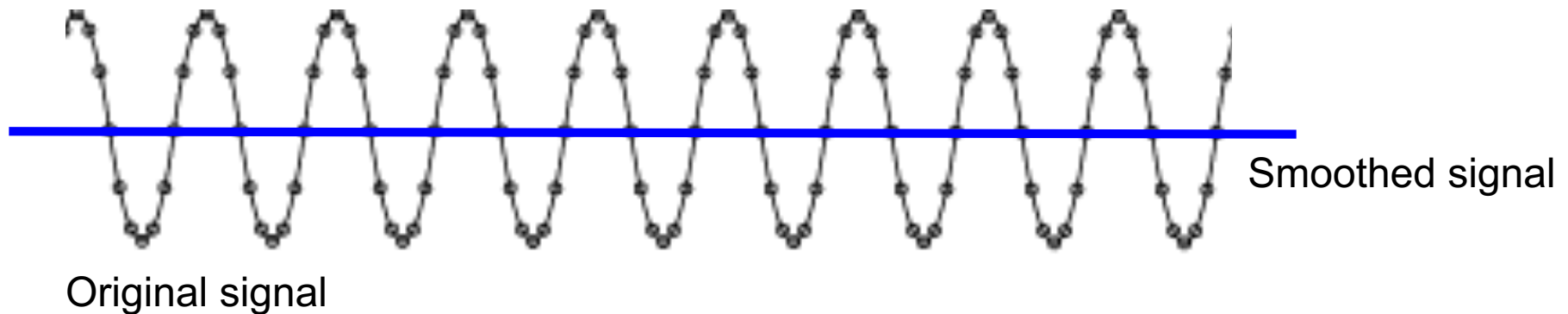


- When **downsampling**, need to compute a summary pixel value from each local area: **Pick one, mean, weighted mean...**



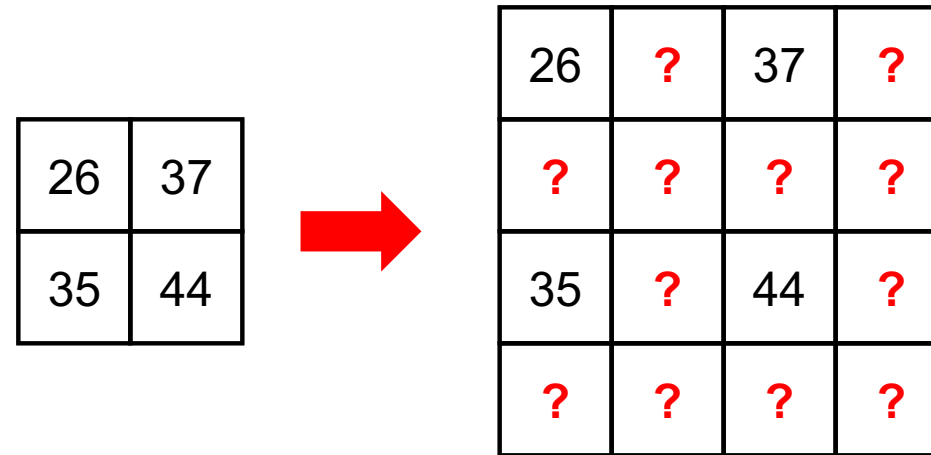
Anti-Aliasing

- Aliasing can be introduced when an image is resampled, if the sampling rate of the new image is less than the Nyquist rate of the original
- Smooth out high frequency signals before sampling so its impossible to “see” the alias





Re-Sampling & Re-Sizing



- When **unsampling**, need to interpolate from the known values to produce an estimate at the unknown pixels
- Average the known values in the local region centred on each unknown pixel, fit some kind of function through known values...



Re-quantisation

Pixel values are integers in a fixed range

Grey level resolution can be dropped by dividing each pixel value by a constant, **but there is a side effect**

Can't increase grey level resolution of a single image

Super-resolution methods exist that combine multiple exposures of the same scene, and so have more than one measurement of each pixel



Key Points

Sampling & quantisation are often under user control during image capture and have a significant effect on the image

Both can be changed to some extent after capture, but the resulting images are often approximations

REMEMBER

Be aware of the Nyquist rule when choosing an image resolution; it can introduce artefacts that are difficult to remove



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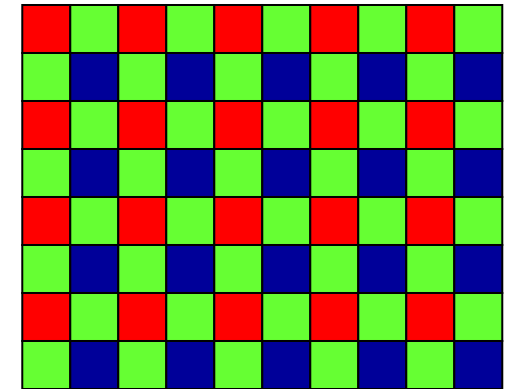
Acquiring & Using Colour Images



Acquiring Colour Images

IMPORTANT

- Each pixel in a grey level image contains one value. Colour requires **three**
- **RGB** model inspired by the retina is an obvious choice for use in cameras
- Some (expensive) cameras use complex optics & 3 CCDs to capture each colour channel independently
- Necessary for some scientific purposes, not for general use
- Single CCD colour cameras use **Bayer pattern**



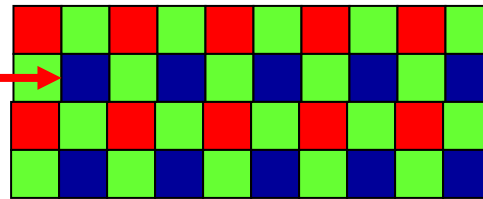
Each CCD cell lies under a red, green or blue filter



Bayer Pattern

One colour value is measured, but two are **estimated** at each pixel.

Here ———→
B is measured, G is the
mean of the 4-neighbours, R
is the mean of the diagonals



More green elements as
our eye is more sensitive
to green light than red or
blue

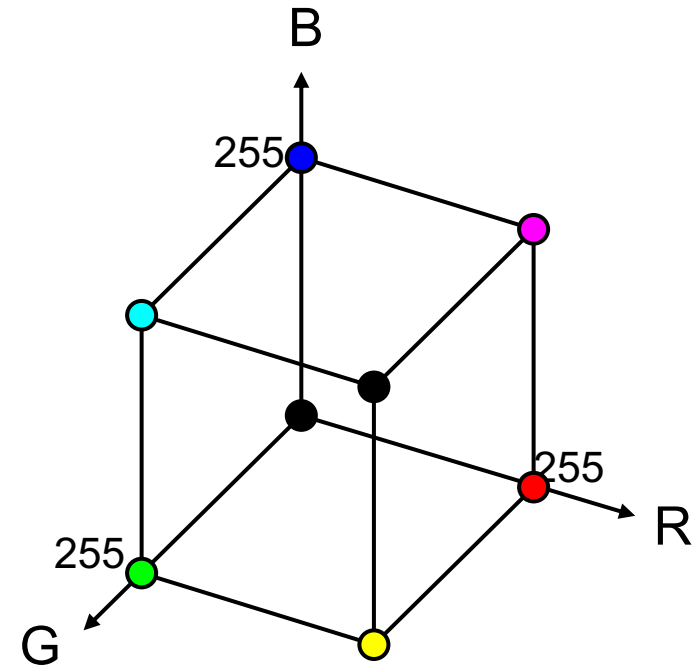
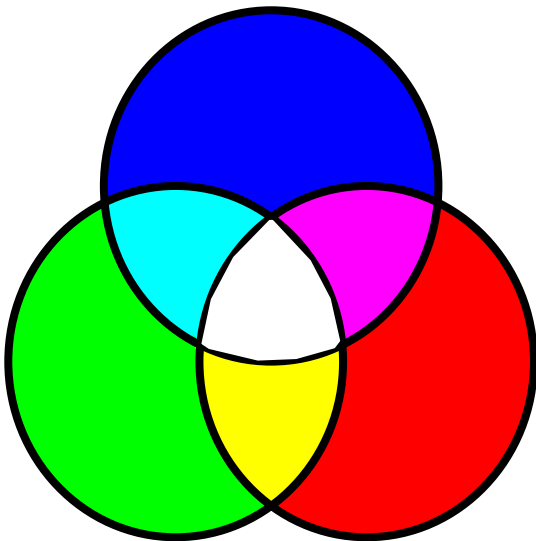
Pattern is repetitive so there are a finite number of sets of filters that
can surround each sensor



Representing Colour: RGB

- Most common starting point
- Retinal cells are sensitive to three primary colours **R**, **G**, **B**
- Light, not pigments

INTERESTING FACT



RGB is **additive**, other colours are made by **mixing** these together



An RGB Example

- Colour images are 3D arrays
- Display software interprets them to produce a colour view

REMEMBER



Colour Image



Red



Green

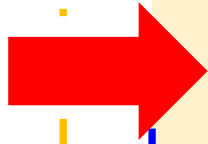


Blue



Colour Vs. Greyscale

Sometimes we want a single value at each pixel



- *Makes processing easier*
- *Reduces the amount of information*
- *Makes some of the theory simpler*

Many image processing methods were developed for single value images

This value is usually the **intensity** or **grey level**

Early CCDs only produced grey level; rare now



- *We can compute the grey value using a simple average of red, green and blue*
- *But our eyes are more sensitive to green light so...*



Colour Vs. Greyscale

We can convert an RGB image to greyscale using:

$$I = 0.30R + 0.59G + 0.11B$$



Average

Original

Weighted



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Colour Spaces

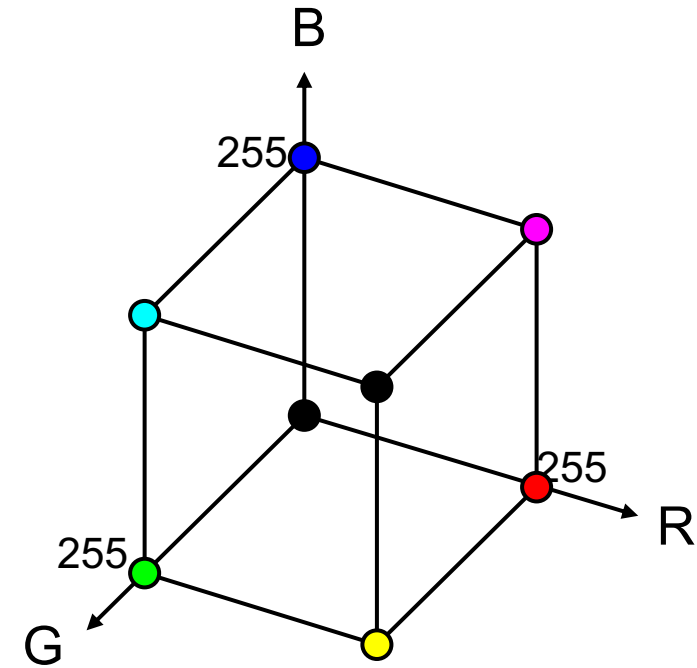
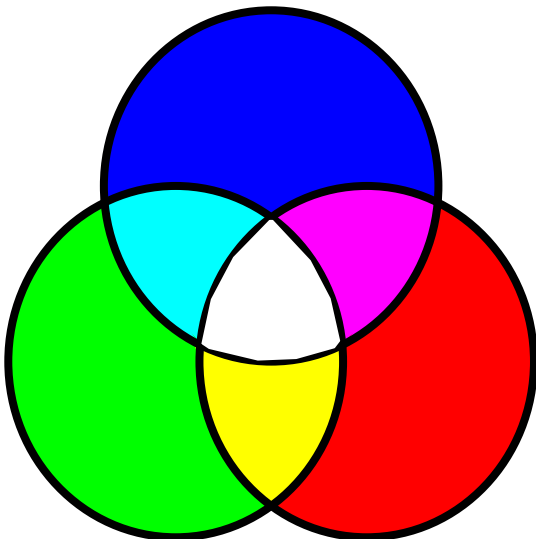


Representing Colour: RGB

Let's Walk DOWN Memory Lane

- Most common starting point
- Retinal cells are sensitive to three primary colours **R**, **G**, **B**
- Light, not pigments

INTERESTING FACT



RGB is **additive**, other colours are made by **mixing** these together



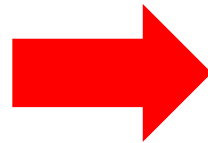
Alternative Colour Spaces

- RGB is OK, but there are other general & application-specific colour spaces...
- If you want to work with plants, you might use just G, or 'greenness':

$$G - (R+B)/2$$



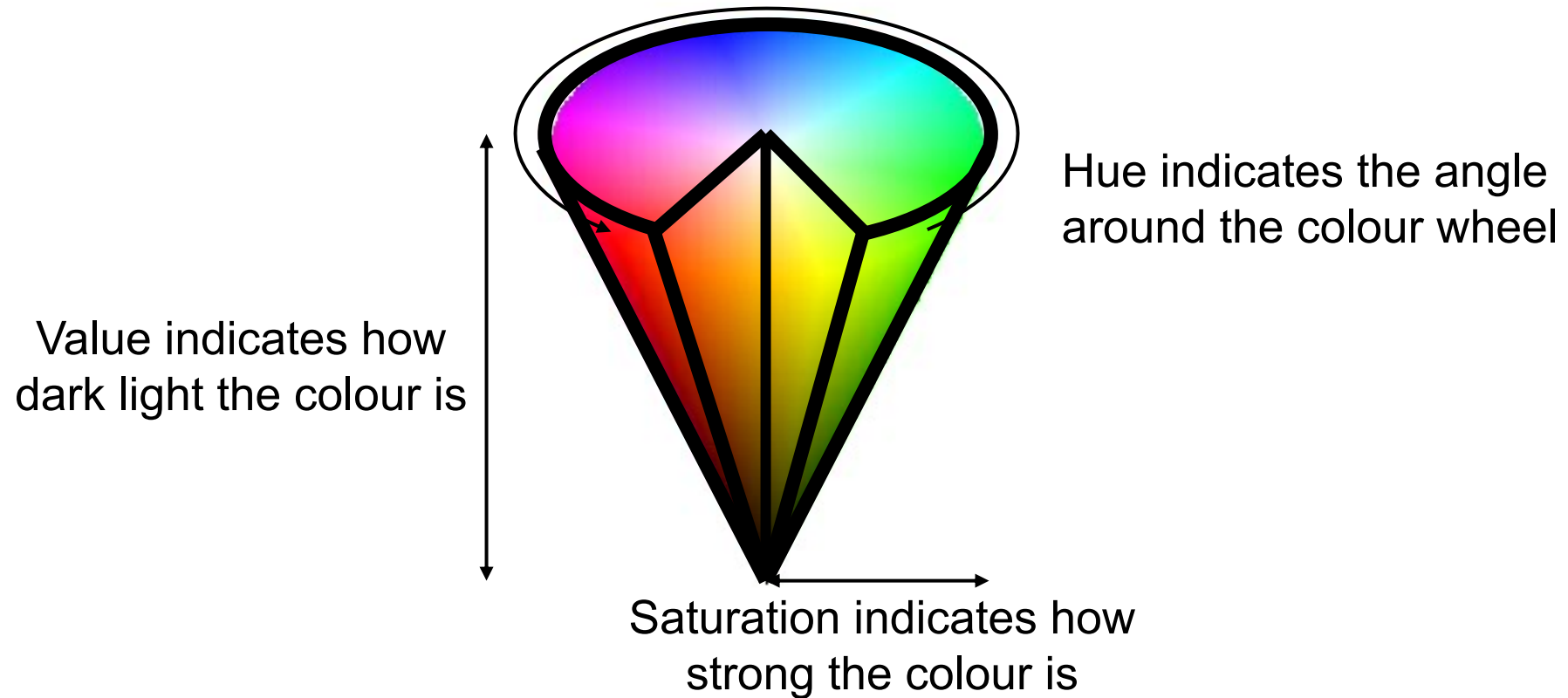
HSV is based on colour rather than light



- *Hue* - what general colour is it
- *Saturation* – how strongly coloured is it
- *Value* – how bright or dark is it



HSV Space





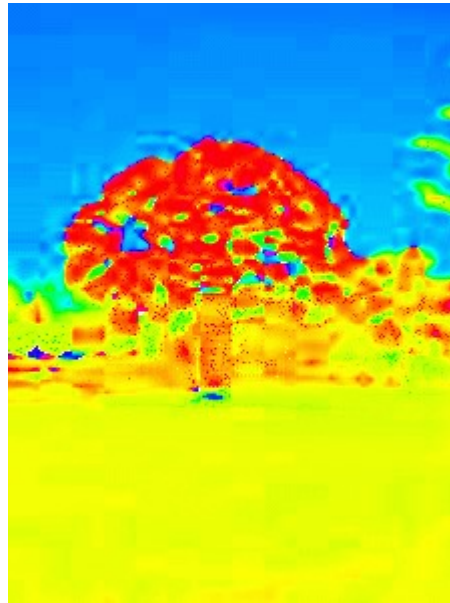
HSV Space

HSV separates colour from intensity (value) making it less sensitive to illumination changes

What makes it DIFFERENT to RGB Space?



Colour Image



Hue

(presented as a colour image)



Saturation

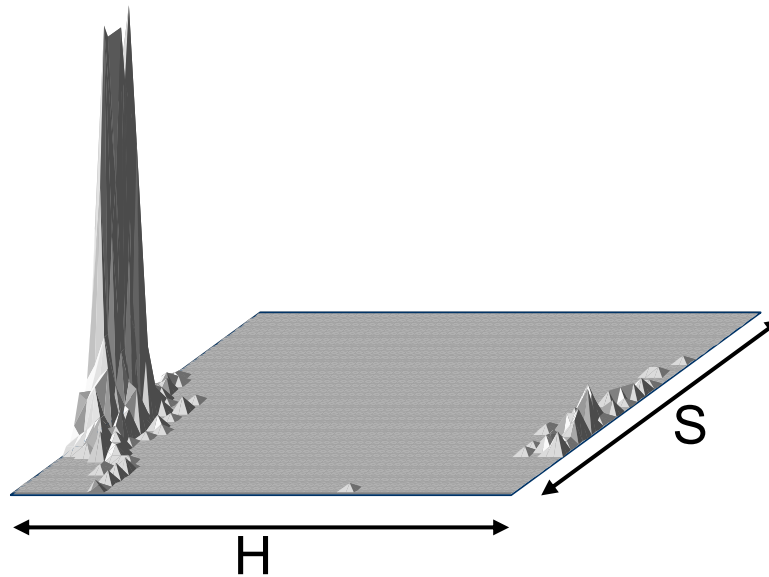


Value



Hue, Saturation & SKIN

- We don't always need all 3 values
- Human skin is tightly clustered in H, S space



Histogram back-projection

ACK: Prof. Tony Pridmore, UNUK



Key Points

Colour space transformation is an example of a point process

Alternative 3D colour spaces exist

REMEMBER

Not all applications require all 3 colour planes to be considered, & application-specific colour spaces exist



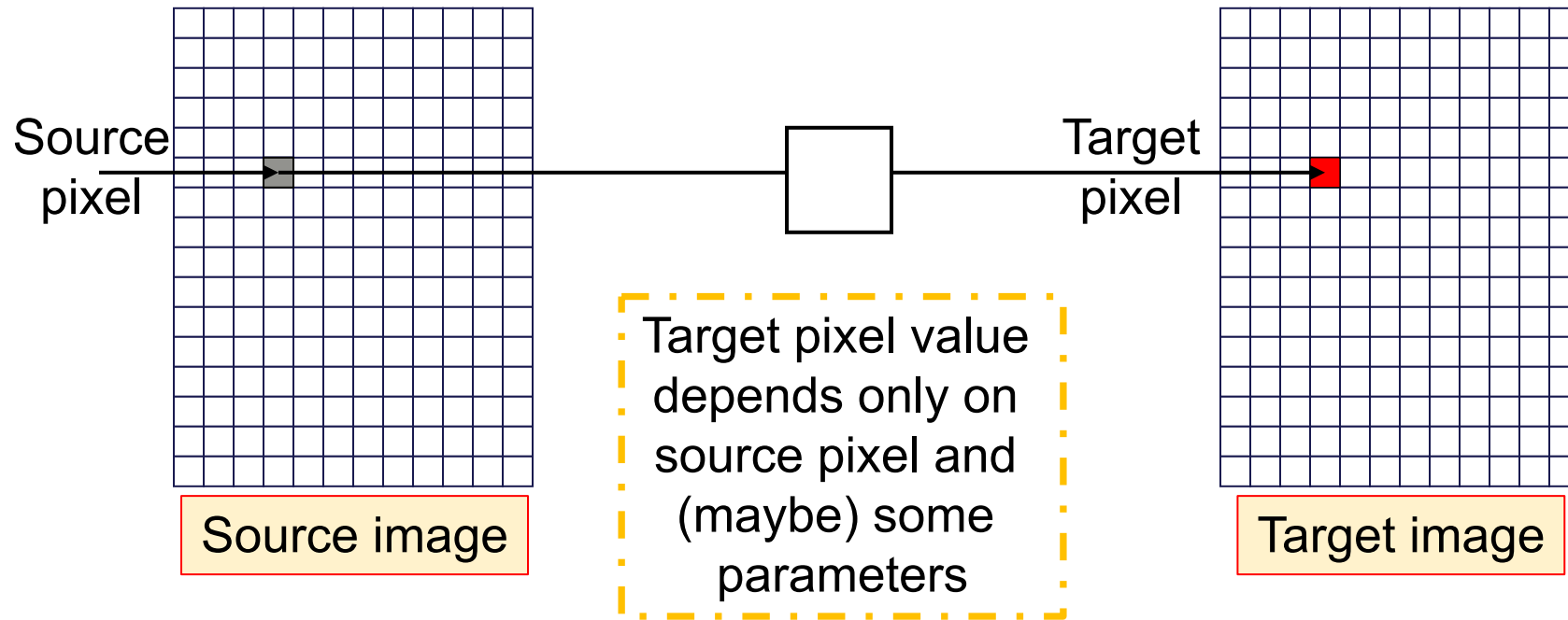
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Intensity Transform



Intensity Transform



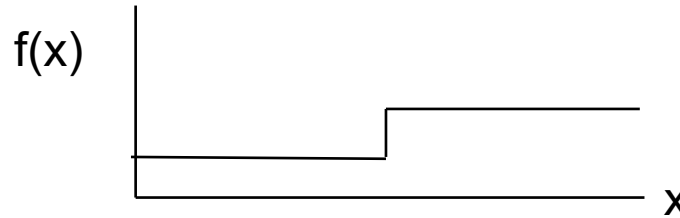


Linear Transforms

2

Commonly used point processes are multiplication by and addition of a constant, i.e.,

$$g(x,y) = a.f(x,y) + b$$

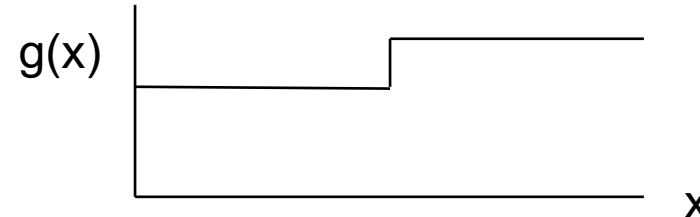
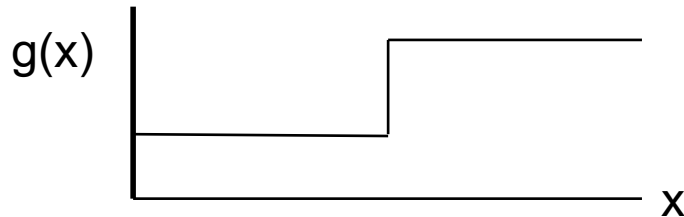


a

Is the *gain*, and controls **contrast**

Is the *bias*, and controls **brightness**

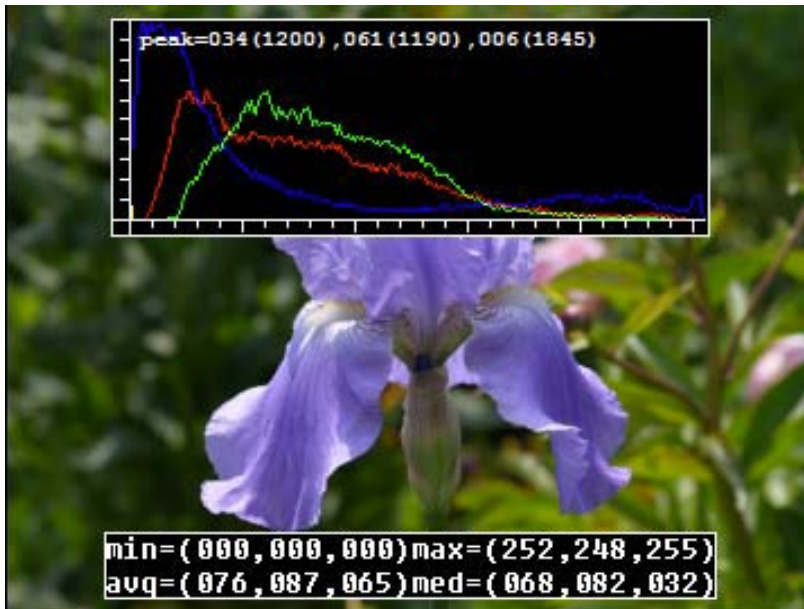
b



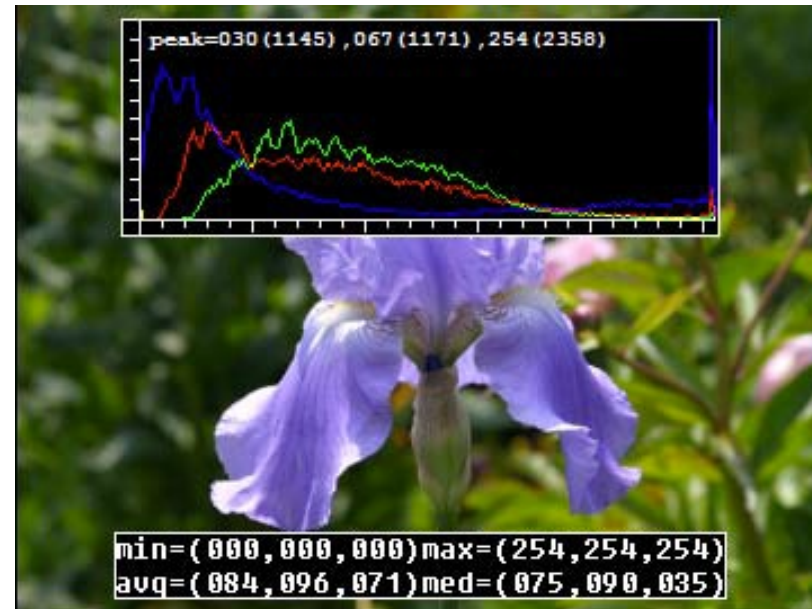


Gain

$$g(x,y) = 1.1 \cdot f(x,y) + 0$$



$f(x,y)$

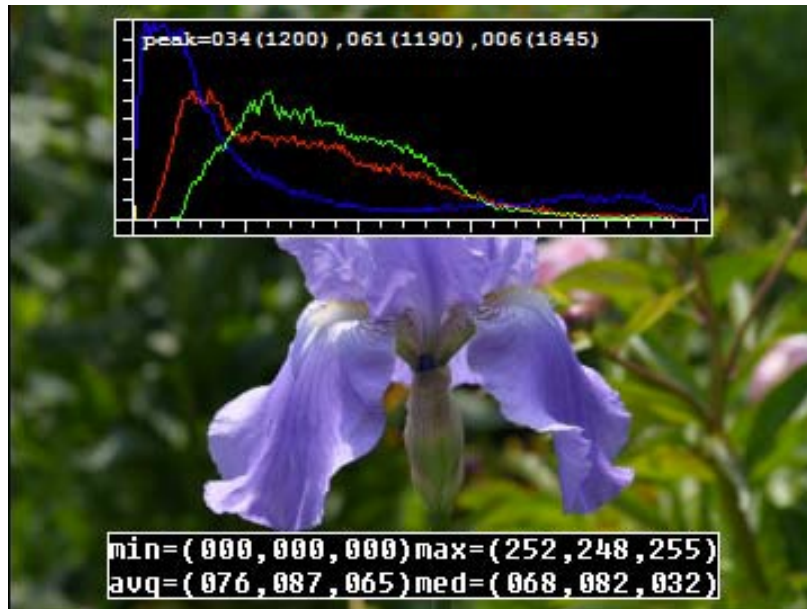


$g(x,y)$

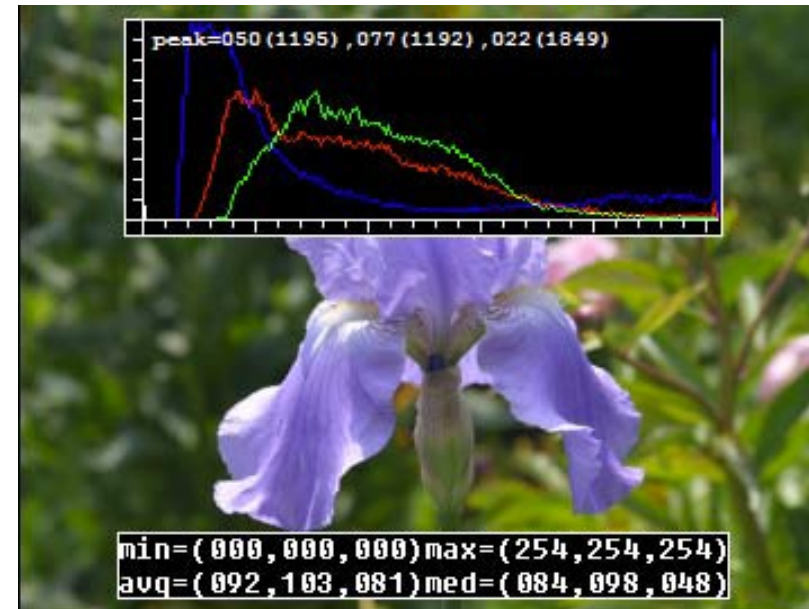


Bias

$$g(x,y) = 1.1 \cdot f(x,y) + 16$$



$f(x,y)$

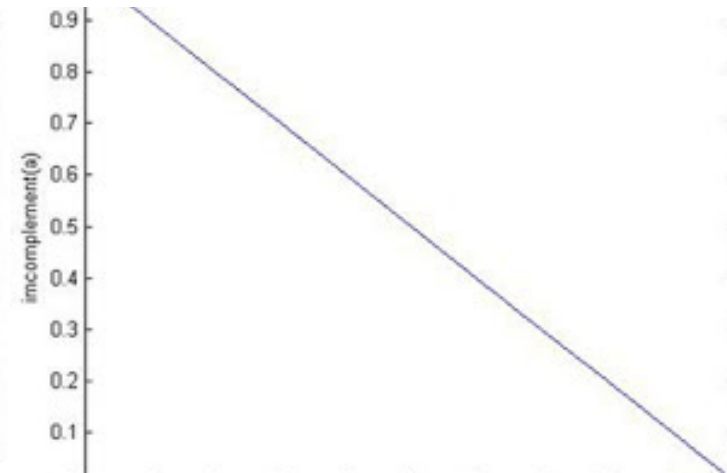


$g(x,y)$



Negation

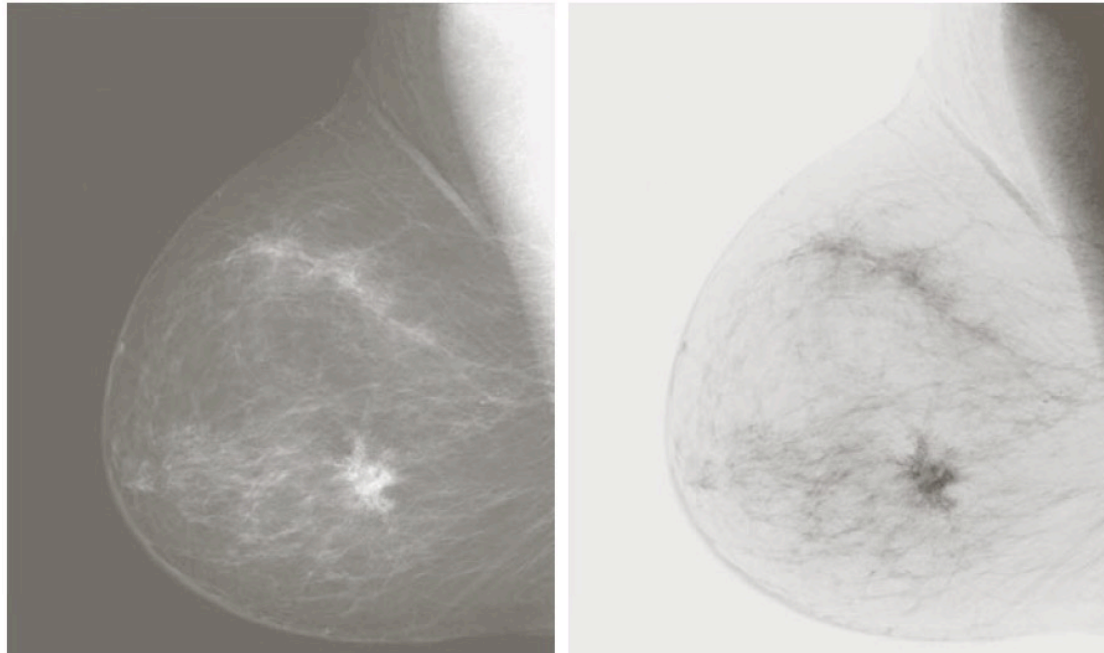
$$\begin{aligned}g(x,y) &= f_{\min} + (f_{\max} - f(x,y)) \\ &= -f(x,y) + (f_{\min} + f_{\max})\end{aligned}$$





Negation

Often used to make fine details more visible, e.g., in **digital mammograms**:

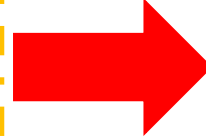




Dynamic Range

- Digital images are sampled – they contain a fixed number of data values
- Digital image representations can only store a fixed number of values

Intensity transform can produce values that are



- *outside that range and so can't be stored*
- *clustered in a small part of that range and so are hard to distinguish*

Some intensity transforms need data in a particular range



Contrast Stretching

Purpose

To convert a source image in which intensities range from min_s to max_s to one in which they range from min_t to max_t

$$g(x,y) = min_t + \left(\frac{f(x,y) - min_s}{(max_s - min_s)} \right) \cdot (max_t - min_t)$$

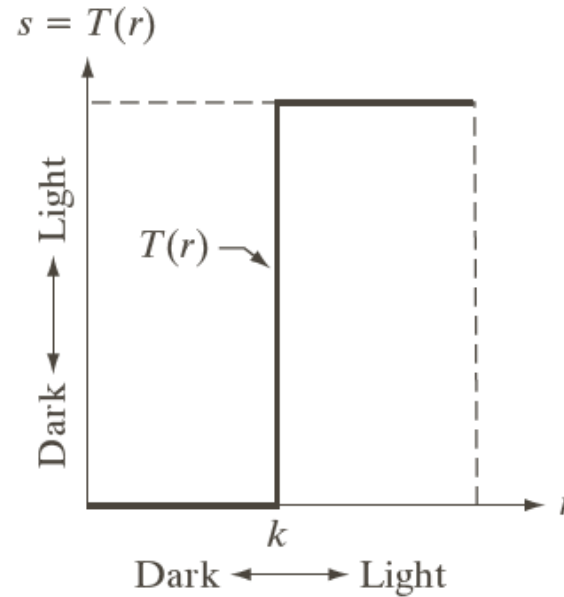


ACK: Prof. Tony Pridmore, UNUK



Non-linear Transform

Thresholding

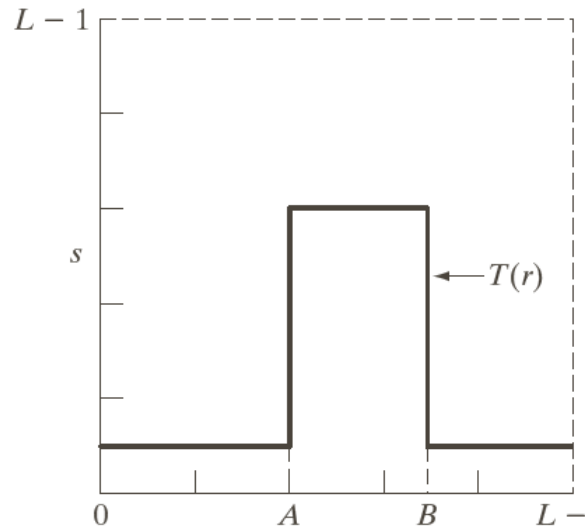


Can be done to highlight grey levels, or as the simplest form of image segmentation

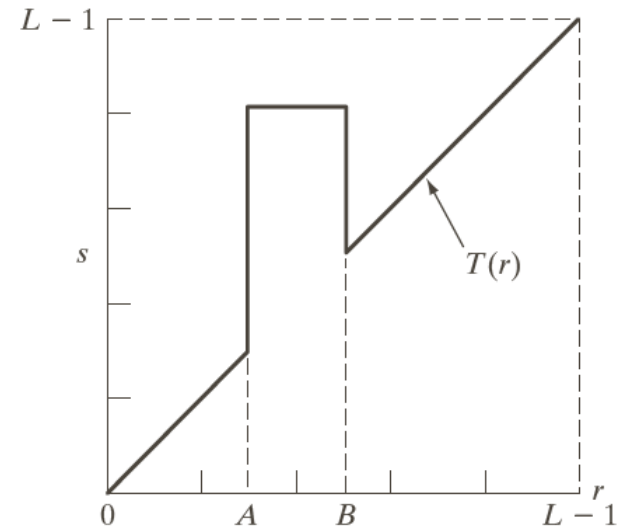


Grey Level Slicing

Highlights a specific range of intensities



Can reduce other grey levels to lower level...



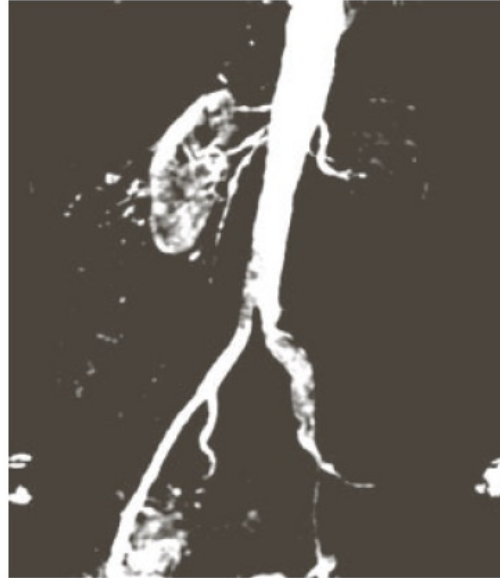
...or preserve them



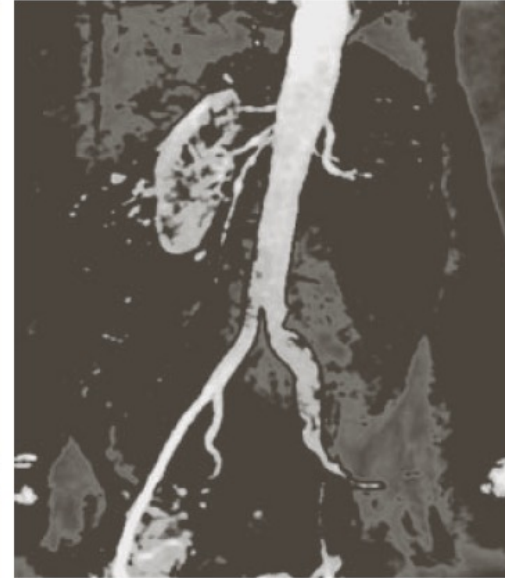
Grey Level Slicing



An aortic
angiogram



High intensities
selected, others
reduced to low
level

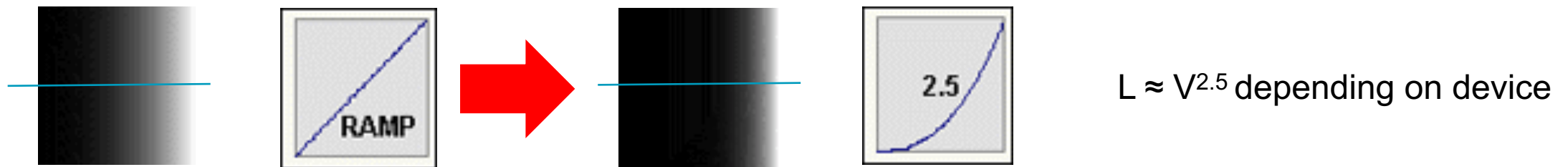


Selected intensities
set to black, others
(blood vessels &
kidneys) preserved



Gamma Correction

- When an image is displayed on a screen, the hardware used effectively applies an intensity transform
- You send a voltage proportional to the intensity of a pixel, the screen displays an intensity that is related to that, but not how you may think





Gamma Correction

- We need to transform the image, so that it generates a voltage which will display what we want

Create a new image in which

$$g(x,y) = f(x,y)^{1/2.5}$$

$$V = g(x,y)$$

$$L = (f(x,y)^{1/2.5})^{2.5} = f(x,y)$$

then

Gamma Correction

Because the equation is usually written

$$g(x,y) = f(x,y)^{\gamma}$$



Key Points

Point processes operate on each pixel independently

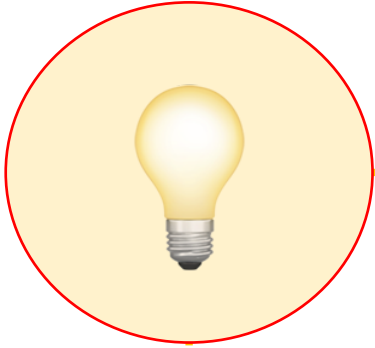
Linear processes change the appearance of the whole image

REMEMBER

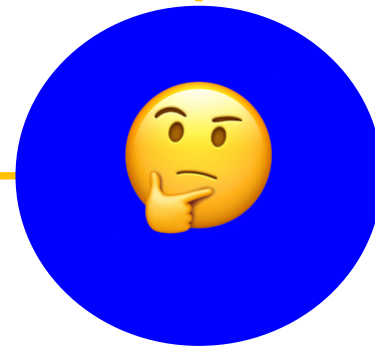
Non-linear processes can differentiate object/image regions



Summary



1. Digital Images Formation
2. Acquiring & Using Colour Images
3. Colour spaces
4. Intensity Transform





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Questions



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NEXT:

**Histogram (Whole
Image Understanding)**